

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 13

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	• Minerals can only be exploited where they were deposited (no choice of location), unlike housing, roads or ports • Land take is larger than the mine void: buildings, access roads, overburden and spoil heaps, buffer zones; open-cast mines take far more land than deep mines	2
01.2	• Dust: water sprays on roads / blasting areas make particles heavier and clump so they settle • Noise: baffle mounds / embankments absorb and deflect; blasting at set, predictable daytime hours; quieter machinery • Turbid water: sedimentation lagoons where water stands until solids settle before discharge • Leachate: collect drainage water and pass through crushed limestone to neutralise acid and immobilise toxic metals • Amenity: landscaping and tree planting to screen the mine; restoration plans agreed as a condition of planning permission	5
01.3	• Landscaping to reduce gradients, compaction, drainage to prevent waterlogging, adding soil / nutrients and planting vegetation whose roots bind the surface • At Aberfan a waterlogged coal-mine spoil heap became unstable and slid into the village • Killing 144 people, mainly children in the primary school	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 139-142; AQA spec 3.2.3.6.	
02.1	• The length of time a pollutant remains in the environment before it is broken down / degraded (chemically or biologically)	1

Question	Marking guidance	Marks
02.2	<i>Any 4 from:</i> • Light: UV light degrades many pesticides (eg neonicotinoids break down fastest in sunlight) so persistence is shorter in bright conditions • Temperature: higher temperatures increase the rate of chemical breakdown / microbial decomposition • Soil microbes / oxygen: aerobic microbial activity breaks pesticides down so persistence is shorter in well-aerated, biologically active soils • pH / presence of other chemicals may increase or decrease the rate of degradation • Adsorption onto clay / organic matter may protect from degradation and increase persistence	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 263-266; AQA spec 3.4.3.2.8.	
03.1	• Bacteria in the gut convert nitrates to nitrites • Nitrites react with haemoglobin so it cannot carry oxygen • Infants are most at risk because they have more of the gut bacteria and their haemoglobin reacts faster; usually from shallow wells receiving arable drainage	3
03.2	• Nitrites may be converted in the gut to nitrosamines which are carcinogenic • But most research has been on rats / the evidence in humans is not conclusive; nitrites are also ingested in processed food so the source cannot be isolated	2
03.3	<i>Any 3 from:</i> • Reduce use of nitrate fertilisers / match application to crop need • Avoid ploughing / applying fertiliser at times of heavy rain (autumn / winter) • Use low-solubility / slow-release fertilisers rather than soluble rapid-release ones • Store manure / slurry where it cannot leach or wash into rivers • Grow crops with lower nitrogen requirements (eg triticale instead of wheat) • Grow legumes with nitrogen-fixing bacteria instead of relying on artificial fertiliser	3
03.4	• Iron(III) sulfate is added to effluent: phosphates are precipitated as insoluble iron phosphate (a fine sediment that can be used as fertiliser) • Lakes: dredging phosphate-rich sediments / adding iron(III) sulfate to reduce phosphate solubility (eg the Norfolk Broads)	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 267-268; AQA spec 3.4.3.2.9.	
04.1	• Most energy comes from non-renewable fossil fuels whose reserves are depleting; the rate of new discoveries has fallen sharply since the 1980s • Use causes unacceptable environmental damage: atmospheric pollution, climate change, oil pollution, habitat loss • Economic / social: depleting resources become more expensive, reducing future affluence; risk of an energy gap if new technologies are not developed in time	3

Question	Marking guidance	Marks
04.2	• Coal mining: habitat loss (especially open-cast), spoil heaps, subsidence from deep mines, acid drainage, dust, noise • Oil and gas extraction: spills from rigs, oil-based drilling mud polluting groundwater and sea, gas flaring releasing SO₂ / CO₂ / smoke • Pipelines causing habitat damage during construction; tanker transport risks • Fuel processing (refineries, coal washing) pollution; embodied energy of equipment and site development for all resources	4
04.3	• New technologies take many years to develop, prove and scale up, and need new infrastructure (grids, storage, vehicles) • Existing cheap fossil fuel technologies make it hard for alternatives to compete until fuel prices rise • By then depletion would reduce supply faster than replacements could be built, causing shortages	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 180-186; AQA spec 3.3.3.	
05.1	• $(15.0 - 2.2) / 15.0 \times 100$ • = 85.3% (accept 85%)	2
05.2	• Added as an anti-knock agent to smooth combustion and reduce engine wear • Lead particulates were released in exhaust gases into the air of cities and inhaled / deposited on soil and food • Lead is a neurotoxin that bioaccumulates; chronic exposure damages the nervous system / brain development in children	3
05.3	<i>Any 2 from:</i> • Replacement of lead water pipes / lead solder with copper / tin-based solder; phosphoric acid dosing of water supplies forming insoluble lead phosphate • Phasing out of lead-based paint / safer paint removal • Lead-free electrical solder • Restrictions on lead fishing weights / shotgun pellets • Industrial emission controls	2
05.4	• Reacts with the lead in old pipes to form an insoluble layer of lead phosphate on the inside of the pipe • This prevents lead dissolving into the water	2
05.5	• Swans / ducks swallow them (mistaking them for snails / grit) and the lead is absorbed, causing lead poisoning	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 275-276; AQA spec 3.4.3.2.11.	
06.1	• Inorganic mercury sank into the sediments / deep water where conditions are anaerobic / low oxygen • It was chemically reduced / converted by (anaerobic) bacteria to methyl mercury	2
06.2	• Methyl mercury is liposoluble so it passes through cell membranes and is stored in fat / not excreted • So it bioaccumulates in organisms and biomagnifies along the food chain (eg in fish, then humans eating fish) • It is a neurotoxin / teratogen causing brain damage and birth abnormalities (eg Minamata)	3

Question	Marking guidance	Marks
06.3	- Mercury is persistent / not degraded; it is an element so cannot be broken down - It remains in the sediments and continues to be released / converted and taken up by the food chain; long-lived fish continue to accumulate it	2
06.4	<i>Any 3 from:</i> - Water current velocity and direction (dispersal) - Properties of the pollutant: solubility, persistence, liposolubility / potential to bioaccumulate - Food chain pathways to humans (fish, shellfish) / the critical group - Sediment type / adsorption onto clay - Volume of the water body / residence time / dilution - Sensitive locations downstream (fisheries, drinking water abstraction)	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 238-242; AQA spec 3.4.1, 3.4.2.	
07.1	- Trade winds blow westwards across the Pacific near the equator, driving a surface current in the same direction - Surface water moving away from South America is replaced by deep, cold water drawn upwards (upwelling) near the coast - The upwelling water is rich in nutrients (nitrates, phosphates) so large algal / phytoplankton blooms occur - These support a rich food web including the anchovy fishery; the water warms as it travels west	4
07.2	- The trade winds weaken or reverse so the surface current slows / reverses and warm water spreads east - Upwelling of nutrient-rich cold water off South America stops - Phytoplankton productivity collapses so the food web / anchovy population collapses; catch fell from about 7.7 to 1.7 million tonnes	3
07.3	<i>Any 2 from:</i> - Heavy rain and floods in the normally dry coastal deserts of Peru / South America - Drought and reduced rainfall in eastern Australia / Indonesia - Droughts in NE Africa, southern Africa, China - Fewer hurricanes in the North Atlantic / fewer cyclones in Japan	2
07.4	Events appear to be occurring more frequently and are driven by wind patterns that warming may alter, but the causes of ENSO are not fully understood and natural fluctuations make trends hard to separate	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 96-99; AQA spec 3.2.1.2, 3.2.2.3.	
08.1	- Habitat change: flooding destroys the previous terrestrial / river habitats but creates a wetland which may be valuable (wetlands are uncommon) - Wildlife barrier: the dam blocks migrating fish (salmon, sturgeon) and prevents recolonisation along the river - Changed river regime downstream: regulated flow removes low-flow periods needed by species such as river turtles nesting on sandbanks and high flows that wash sediment from gravel spawning beds - Sedimentation: sediment settles in the reservoir instead of fertilising the floodplain and building river banks / coastlines downstream, increasing erosion - Microclimate: water's high heat capacity reduces temperature fluctuations; higher wind speeds; more evaporation, humidity, cloud and precipitation downwind	6

Question	Marking guidance	Marks
08.2	• Vegetation / dead organic matter under water decays anaerobically • Releasing methane, a powerful greenhouse gas	2
08.3	• After heavy rain surplus water is stored, reducing peak flow / flooding downstream • In dry periods the dam releases water to maintain river levels / supplies	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 117; AQA spec 3.2.2.1.	
09.1	• Trays of the same soil (texture, compaction, organic matter, depth) with different percentage vegetation cover (eg 0, 25, 50, 75, 100% grass), replicated • Same slope / gradient; simulated rain from a standard height at a standard rate and drop size (eg watering can / sprinkler) for a standard time • Collect the soil splashed / washed off (eg onto a sheet or into a trough at the base), dry and weigh it • Repeat each treatment several times and calculate means (and standard deviation) • Plot soil loss against cover; statistical test (Spearman's rank / t-test) for significance	5
09.2	<i>Any 3 from:</i> • Rainfall intensity / raindrop size / height • Soil texture, compaction, organic matter, depth • Permeability of the material beneath • Gradient • Root binding	3
09.3	• All variables except the independent variable can be standardised in the laboratory; in the field uncontrollable variables (weather) must be measured instead • Laboratory conditions may not represent natural conditions so results may not apply directly to real ecosystems	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-393, 419-420; AQA spec 3.7.1, 3.7.2.2.	

Question	Marking guidance	Marks
10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Properties of renewables: intermittency, predictability, low energy density, storage, applicability, locational constraints, impacts, technology level, economics • Wind: large offshore resource in shallow seas, falling costs, HAWT / VAWT / direct drive developments; intermittent, needs storage and grid; land-use conflicts • Solar: PV costs falling rapidly, roof space, but low winter insolation in the UK; CSP unsuitable; heat pumps and photothermal for heating • HEP: most large sites used; low-head, helical and micro-hydro extend use; pumped storage vital • Tidal: large range (Severn, Mersey), predictable; barrages costly with high impact, lagoons and in-stream turbines lower impact • Wave: long Atlantic fetch, but technology immature • Biofuels: wastes, anaerobic digestion, energy crops vs food / habitat • Geothermal: limited to low-temperature schemes / heat pumps • Systems issues: storage (pumped HEP, batteries, hydrogen, P2G, V2G), grid management, diversity of sources, energy conservation reducing demand, electrification of transport • Judgement: renewables could supply most electricity with storage and a diverse mix, but heat and transport are harder; transition needs investment and lifestyle change	9
10.2	• Eight times (energy proportional to velocity cubed)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 196-217; AQA spec 3.3.4.1.	

Question	Marking guidance	Marks
11.1	Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: toxicity (LD50, MDAF, MDNF), specificity, persistence, liposolubility/water solubility, mobility, mode of action (contact/systemic), synergism • Bioaccumulation and biomagnification explained by persistence + liposolubility; DDT and birds of prey as a case study • Comparison of organochlorines, organophosphates, pyrethroids, neonicotinoids with their relative advantages and disadvantages • Reducing impact: restrictions/bans, choice of non-persistent and specific pesticides, systemic pesticides and seed coatings, application timing (night, still days, not in flower), droplet size • Alternatives: cultural control (rotation, barrier/companion crops), biological control, predator habitats, sterile males, pheromone traps, genetic resistance, GM crops, integrated pest management • Trade-offs: yield, cost, human health (organophosphates), resistance evolution, pollinator services; risk:benefit; sustainability and the circular economy • Judgement: no pesticide is perfectly specific; combination of property-based selection and integrated control is most sustainable Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-240, 262-266, 314-320; AQA spec 3.4.3.2.8, 3.5.1.1, 3.5.1.5.	25

Question	Marking guidance	Marks
11.2	Discuss how knowledge of the properties of pollutants can be used to reduce the severity of their impact on the environment. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Properties: state of matter, density, persistence, toxicity, specificity, reactivity, adsorption, solubility, bioaccumulation / biomagnification, synergism, mutagenic / carcinogenic / teratogenic action, mobility • How properties predict behaviour: dispersal (gases vs solids, density - Bhopal), how far and how long (persistence, mobility), where they concentrate (adsorption on sediments, food chains) • Environmental features interacting with properties: temperature inversions, currents, pH, oxygen, light, adsorbent materials • Control based on properties: persistent / liposoluble pesticides banned (organochlorines) and replaced by non-persistent / specific ones; systemic pesticides and seed coatings • Heavy metals: lead uses discontinued, storage at high pH to reduce solubility; mercury chemical form; radioactive waste categorised by half-life / activity and contained (encapsulation, sealed storage) • Reactive primary pollutants controlled to prevent secondary pollutants (catalytic converters, FGD, urea sprays); oil properties (floats, insoluble) exploited by booms and skimmers; density in cyclone separators • Principles: CPA and CGM based on properties; emission location / timing; ALARA and BATNEEC; polluter pays; precautionary principle • New technologies: satellite oil-spill monitoring, GPS, polymer adsorption, phytoremediation, bioremediation • Evaluation: prevention better than cure; limits of remediation; cost-effectiveness; contrast success of ozone (Montreal) with climate change; judgement Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-245, 262-266, 275-285, 295-304; AQA spec 3.4.1-3.4.3.	25